

Master EQ

A mastering equaliser with stepped controls and three dynamic bands.



Overview

Master EQ is built for the last stage of a mix. It has low and high cut filters, a low shelf, three bell bands, a high shelf and an AIR band. The controls move in steps, as on classic mastering equalisers, so settings are easy to note down and recall.

The three bells, DYN 1 - 3, are dynamic. Besides a fixed GAIN, each has a COMP knob: turned left it pulls the band down when that part of the music gets loud (taming harshness or boom only when it happens); turned right it lifts the band as it gets busy. In the middle the band is a plain static bell.

Every band can work on the stereo pair, on the mid (the centre of the image) or on the side (the edges), so you can, for example, tighten the lows in the sides only or add air to the edges.

Reach for it for

- Final tonal balance on a mix or master

- Taming harshness or sibilance only when it flares up (DYN with COMP)
- Controlling a boomy low end that comes and goes
- Mid/side work: centring the lows, widening the top
- Adding gentle air above the audio band

Controls

Drag a knob up or down to turn it (hold Shift for fine moves), use the mouse wheel, or double-click to reset it. Gains move in 0.5 dB steps; FREQ knobs step through the standard series (100, 120, 150, 180, 220, 270, 330, 390, 470, 560, 680, 820 and so on), and Q steps through 0.4, 0.5, 0.7, 1, 1.4, 2, 2.8 and 4. The value shows while you touch a knob.

The header

Colour swatch	Opens the list of colour themes for the panel.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
EQ IN	Switches the shelves, the bells, AIR and the dynamic action in or out (the change glides rather than jumps). The cut filters stay in. While it is out, the display reads EQ OUT.
PRESETS	Opens the list of factory presets (clicking the display does the same). A preset sets the EQ and dynamics sections and switches EQ IN on; TRIM, OUTPUT and the OUT stage are left as they are. The last entry, Reset (flat), sets every band flat.
Left / right arrows	Step to the previous or next preset.

The display

Shows the summed EQ curve from 20 Hz to 20 kHz over a range of +/-8 dB, moving with the dynamic bands as they work. The faint vertical lines mark the centres of DYN 1 - 3.

IN + FILTERS

TRIM	Input level before anything else, from -24 to +12 dB. Double-click to reset it.
GUARD (click TRIM)	A single click on the TRIM knob switches GUARD on or off; the small light beside the knob shows it (dark = off, red = on, bright = working). GUARD is on by default. It gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources.
HPF	A 12 dB/oct high-pass filter, stepping from 20 to 300 Hz. Fully left (20) is OFF - out of circuit.
LPF	A 12 dB/oct low-pass filter, stepping from 4 to 20 kHz. Fully right (20 kHz) is OFF.
TIMING	FAST, MED or SLOW: the attack and release of all three dynamic bands (2 / 60 ms, 10 / 150 ms, 30 / 400 ms). FAST follows transients and sibilance; SLOW rides the broader balance.

LOW

30 / 60 / 100 / 200	The low shelf frequency.
GAIN	-6 to +6 dB in 0.5 dB steps.
BROAD / TIGHT	BROAD: a gentle, wide slope. TIGHT: a steeper shelf that changes less above the frequency.
ST / M / S	What the band works on: the stereo pair, the mid only or the side only.

DYN 1, DYN 2, DYN 3

Three identical dynamic bells. The rows are named once on the left: FREQ, GAIN, Q and DYN. The light next to each band's name shows when its dynamic part is working.

FREQ	The centre of the band, 20 Hz - 20 kHz (120 Hz, 1 kHz and 4 kHz by default).
GAIN	A fixed boost or cut, -6 to +6 dB.
Q	The width, 0.4 - 4. Higher is narrower.
DYN (COMP / EXP)	The dynamic part, from COMP 10 on the left through STATIC in the middle to EXP 10 on the right. Towards COMP the band is pulled down when it gets loud, up to 12 dB; towards EXP it is lifted as it gets busy, up to 6 dB. Turning further makes it react at lower levels and more strongly, with a soft, gradual knee. Each band listens only to its own range.
ST / M / S	What the band works on: the stereo pair, the mid only or the side only.

HIGH

2k / 4k / 8k / 12k	The high shelf frequency.
GAIN	-6 to +6 dB in 0.5 dB steps.
BROAD / TIGHT	BROAD: a gentle, wide slope. TIGHT: a steeper shelf.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper that holds the peaks at -0.3 dBFS (no latency), or a lookahead limiter that holds them at -0.3 dBFS (3 ms latency, compensated by REAPER). The CLIP key flashes red while the clipper is working.
ST / M / S	What the band works on: the stereo pair, the mid only or the side only.

AIR

10k / 20k / 40k	Where the broad air lift is centred.
GAIN	0 to +6 dB in 0.5 dB steps.

OUTPUT	The output level, -12 to +12 dB.
Level lights	Ten lights showing the output level, from -30 dBFS up to just under 0 (green, then amber, then red).
ST / M / S	What the band works on: the stereo pair, the mid only or the side only.

Presets

Gentle Polish	A touch of low weight and air, with the low mids, upper mids and harshness gently held in check by DYN 1 - 3.
Warm & Full	A 100 Hz lift, dynamic control of the low mids and the 5 kHz area, and a slightly softer top.
Bright & Open	A small low-mid cut, dynamic control at 2.5 and 6.5 kHz, and more high shelf and air.
Low-End Control	HPF at 30 Hz, the lows cut in the sides only, dynamic control at 110 Hz (mid) and 300 Hz, FAST timing.
Vocal Forward	Presence lifted and mud cut in the mid, sibilance tamed dynamically in the mid, a little air.
De-Harsh	Dynamic cuts at 3.2 kHz and 8 kHz only, working when the harshness flares up.
Wide Air	Lows cut and highs and air lifted in the sides only, keeping the lows centred.
Loud Modern	HPF at 30 Hz, a tight 60 Hz lift with dynamic control at 90 Hz and 3.5 kHz, a low-mid cut, a tight 12 kHz lift and air; FAST timing.
Vintage Tape	Round lows, a softened top and the LPF at 16 kHz.
Acoustic / Classical	Barely there: a hint of low and high lift, gentle low-mid control, a little 40 kHz air, SLOW timing.

Reset (flat)

Every band flat, filters off, dynamic bands static.

Tips

- Use DYN instead of GAIN when a problem comes and goes: a harsh vocal at 3 kHz only on the loud notes, or a bass note that booms only now and then.
- Switch a band to S to work on the edges only - cutting the lows in the sides tightens the bass without making it thinner in the centre.
- Compare with EQ IN on and off; the HPF and LPF stay in either way, so you hear only what the bands are doing.
- The stepped controls make settings easy to note down: write them down to recall a master later.

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